

Target and stitched multiple-shooting estimate

actual common support: 19.009–23.959 s

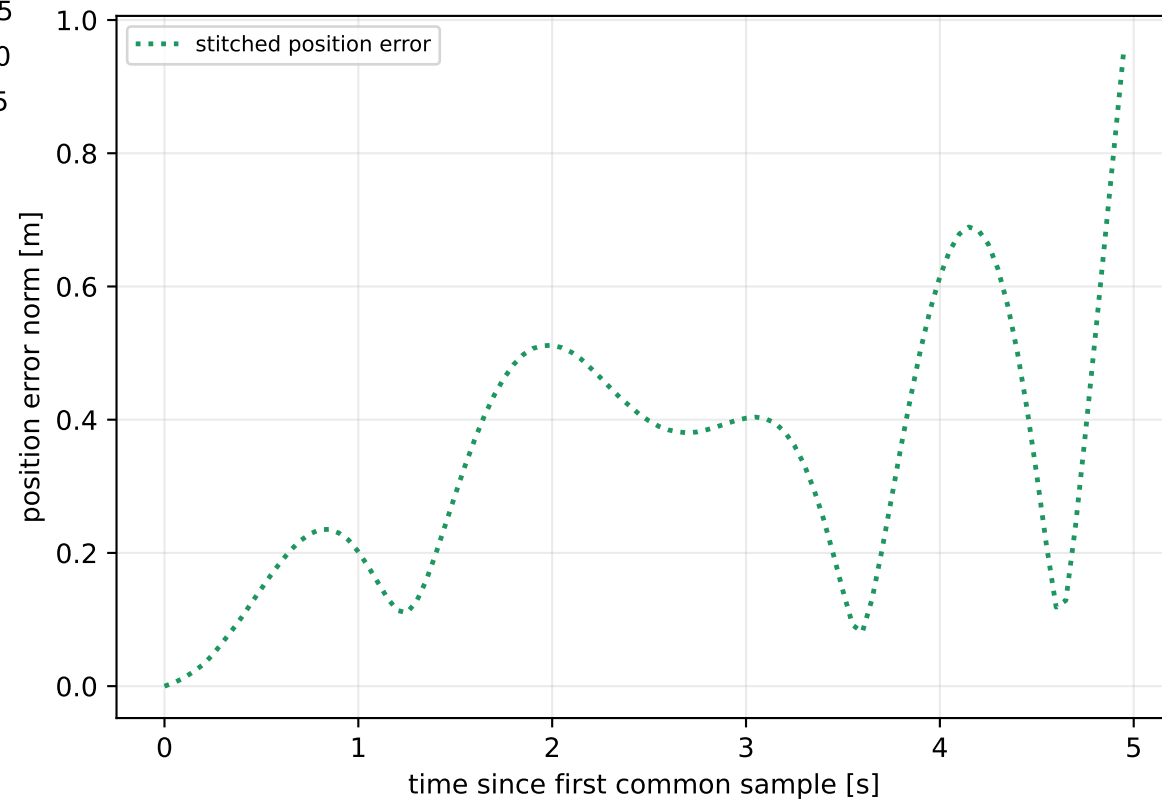
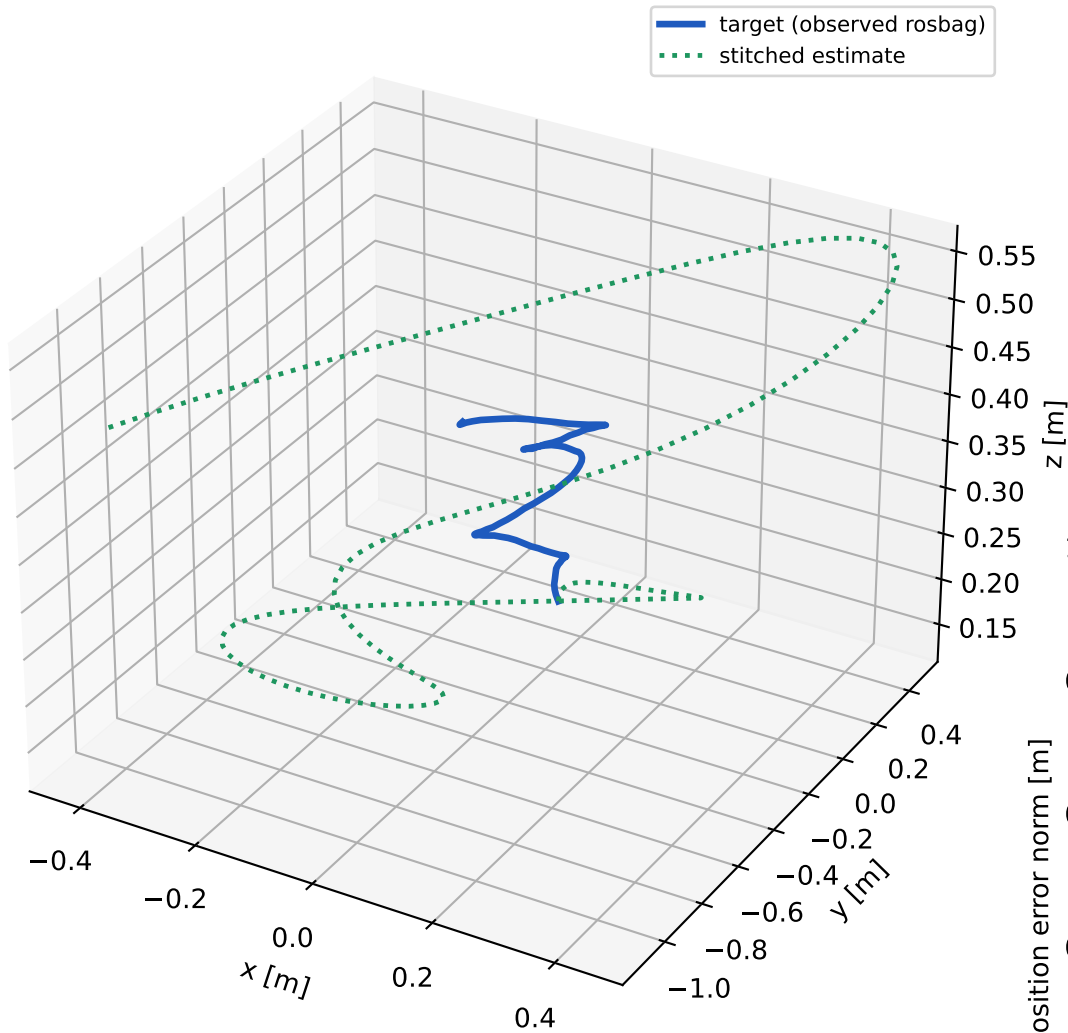
selected command delay: 0.16 s

metric	stitched
position RMSE [m]	0.38844
orientation RMSE [deg]	25.801
velocity RMSE [m/s]	1.1501
angular velocity RMSE [rad/s]	0.71136
specific force RMSE [m/s ²]	1.1677

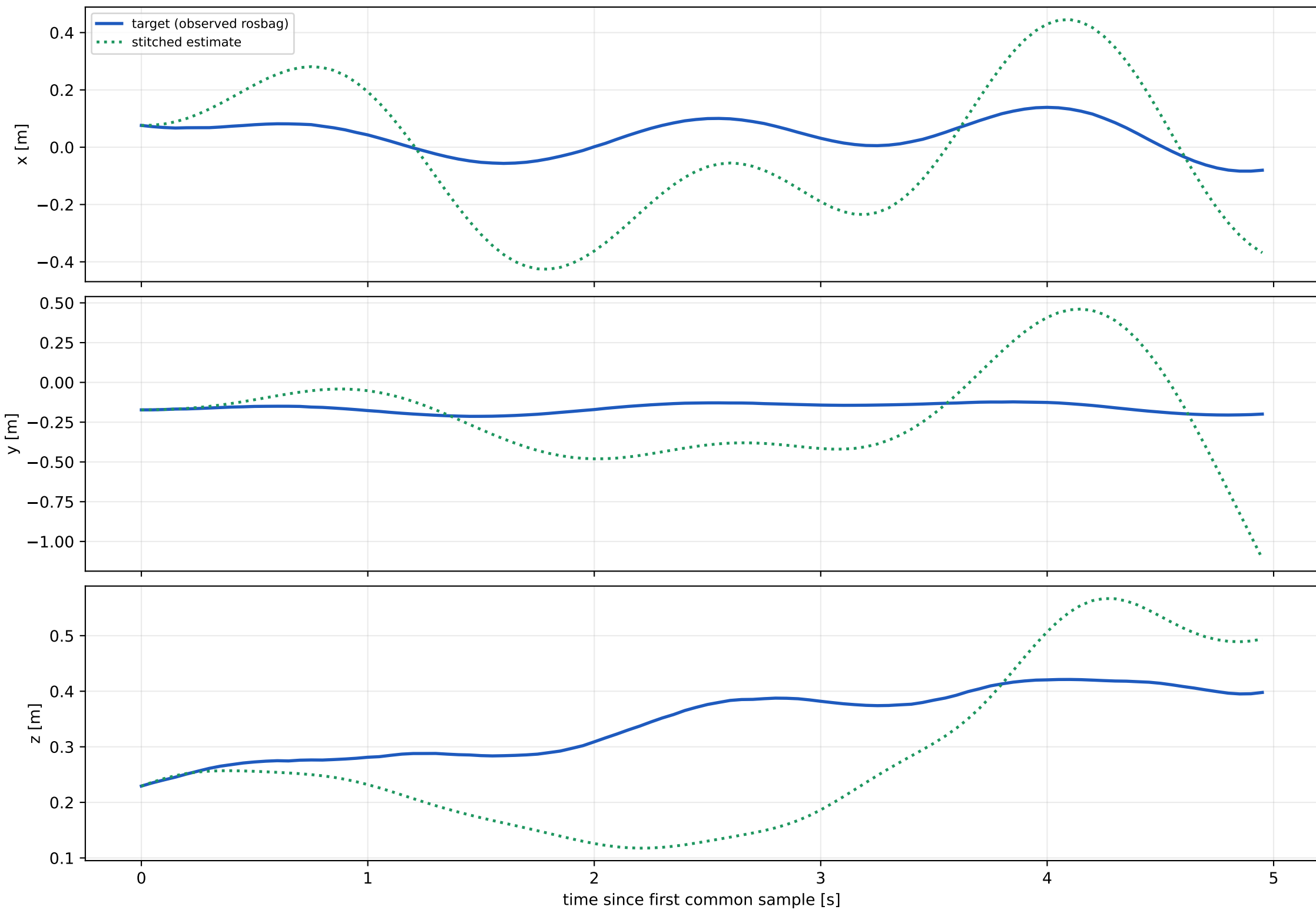
continuity max / tolerance:
0.0001881 / 0.001 (converged)

Velocity and IMU metrics are diagnostic only;
the fit uses sensor position and orientation.

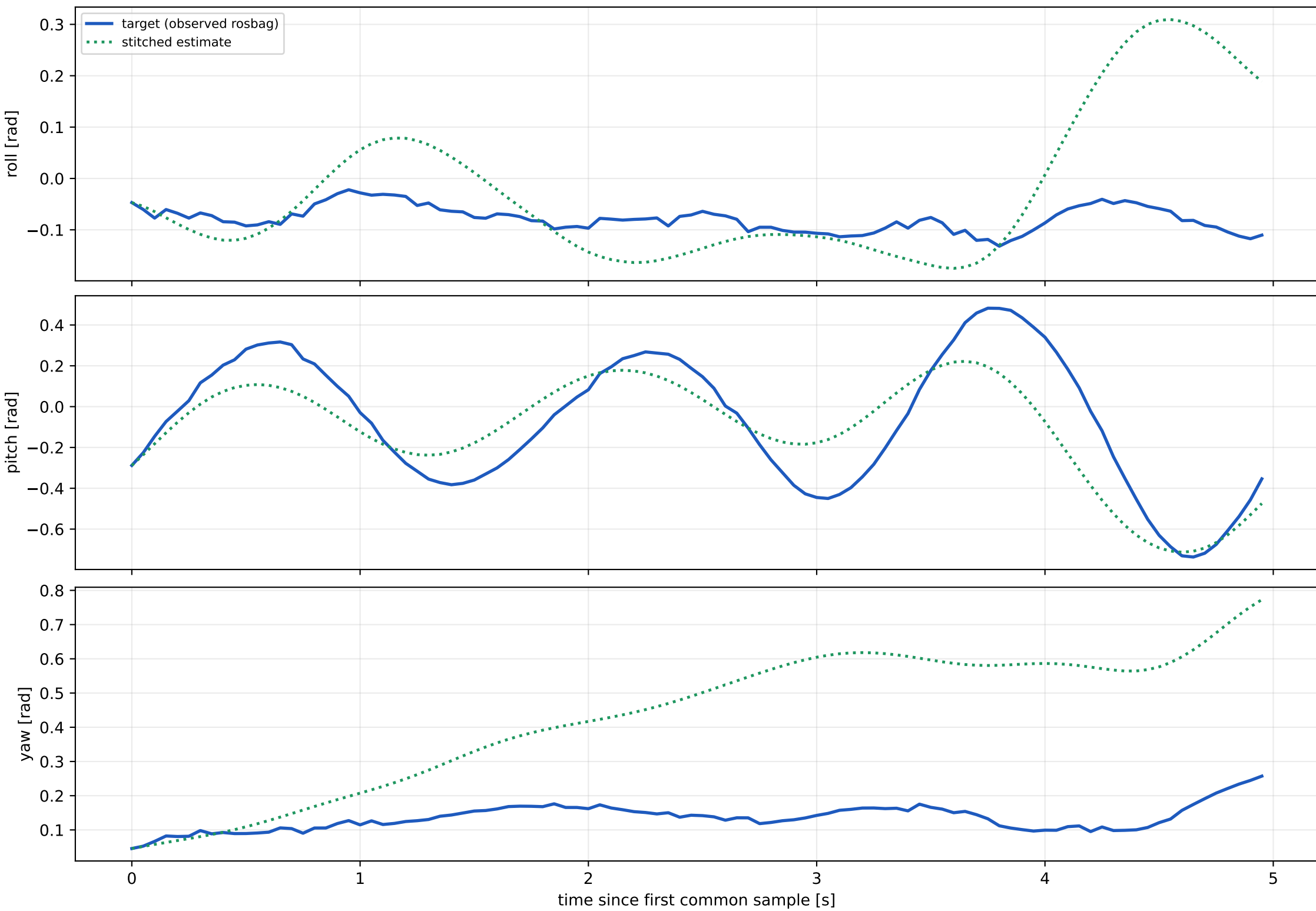
Stitched trajectory



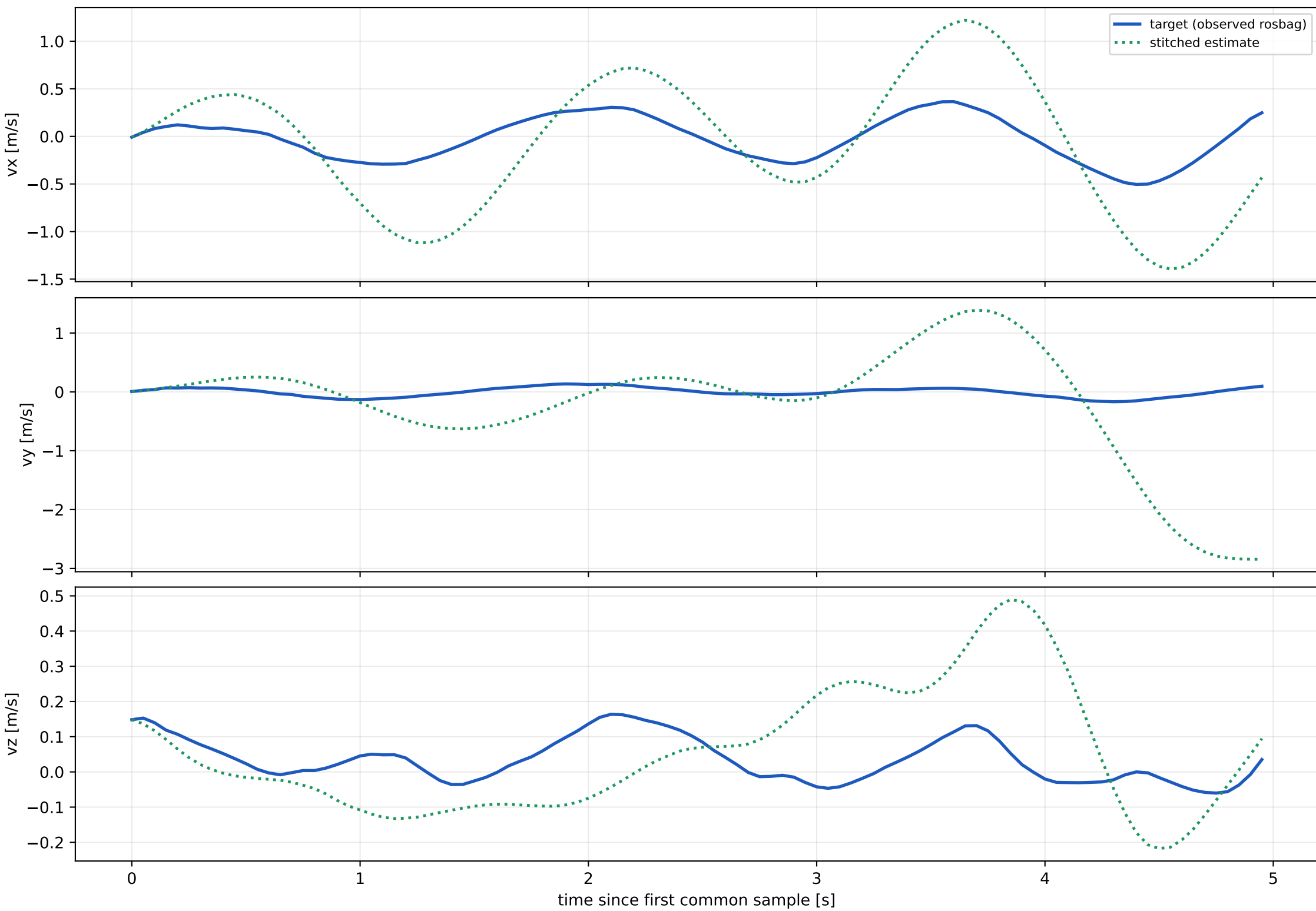
Sensor position



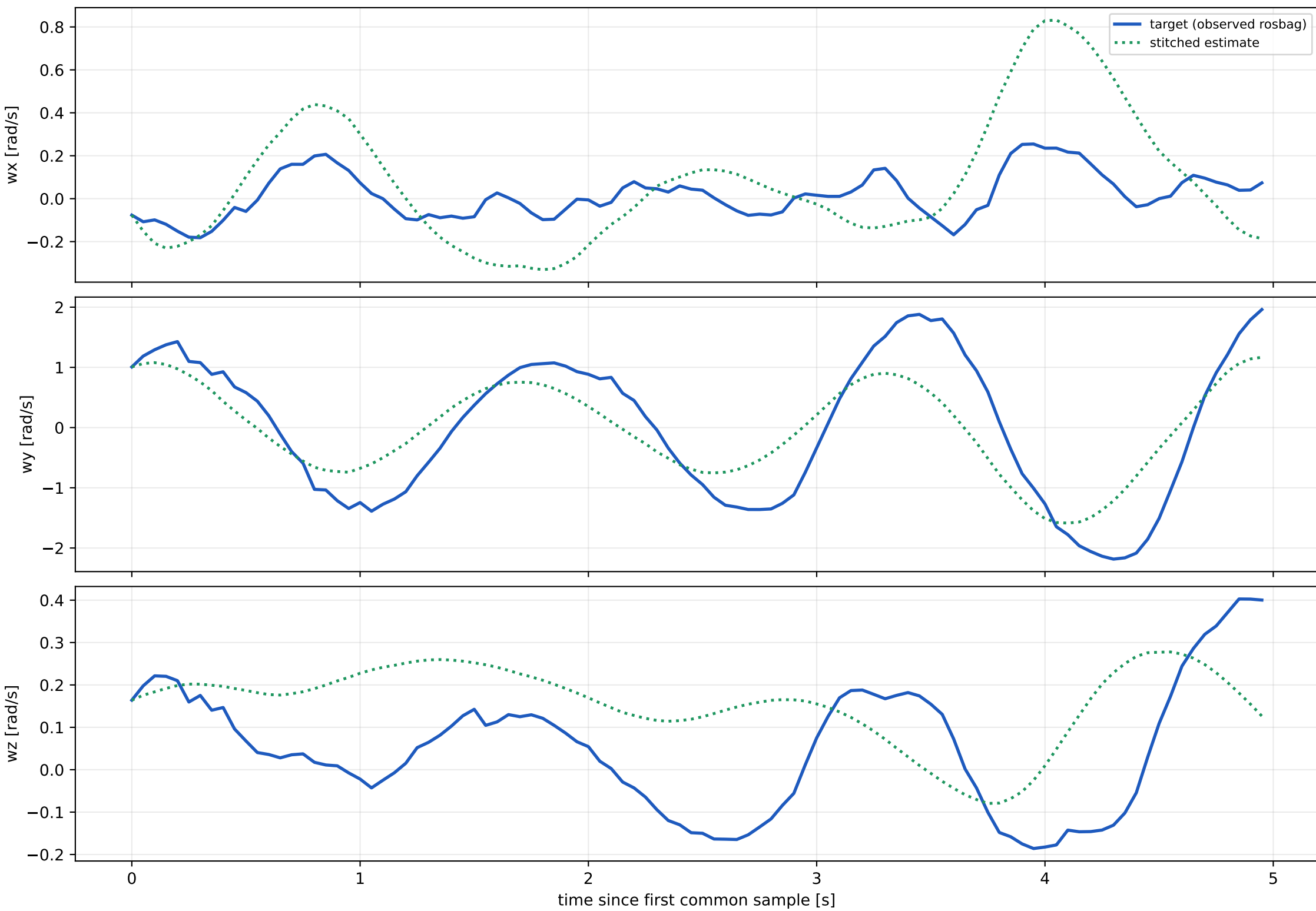
Sensor orientation (display only; fit uses SO(3) log)



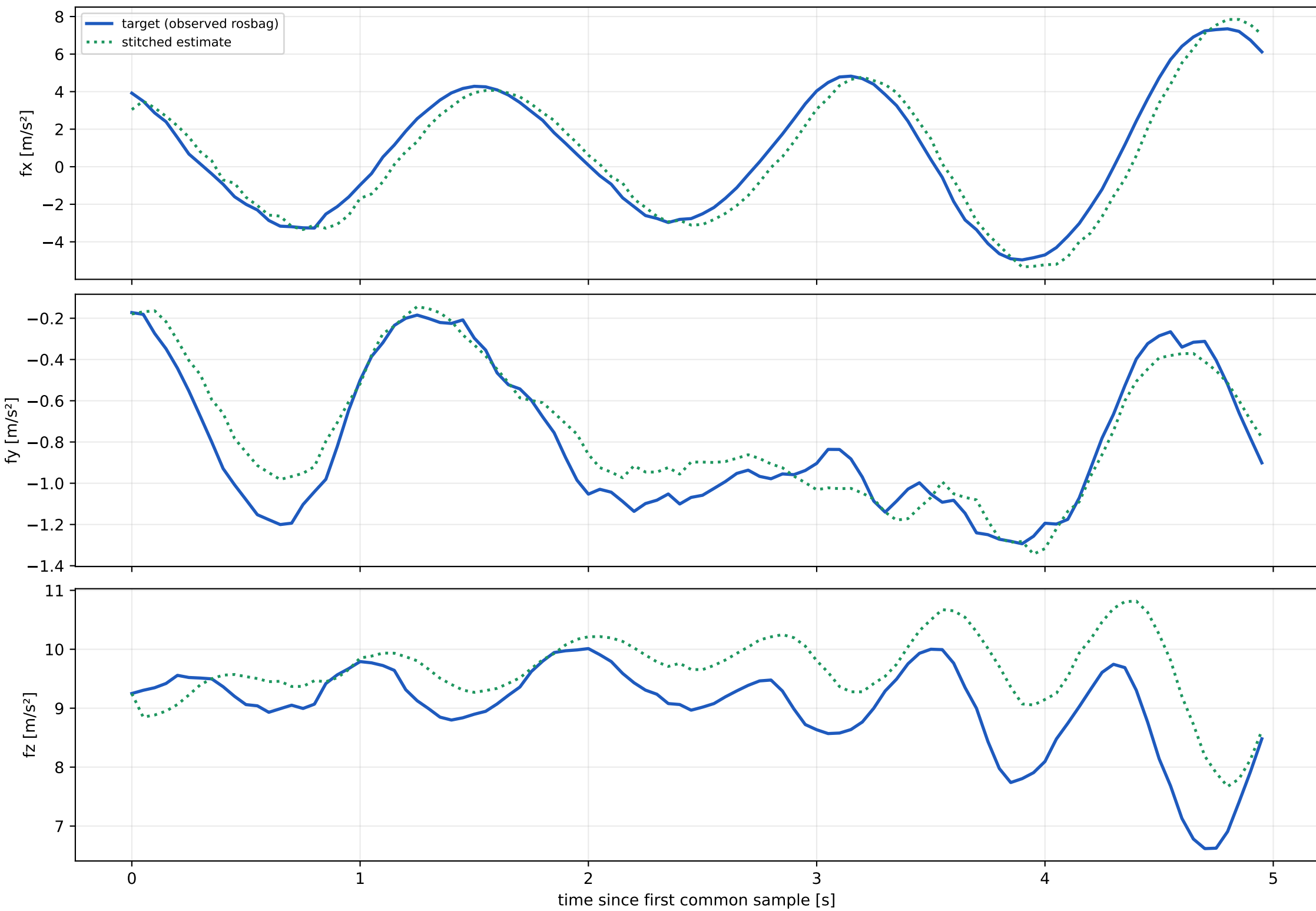
World-frame sensor velocity (diagnostic; excluded from fit)



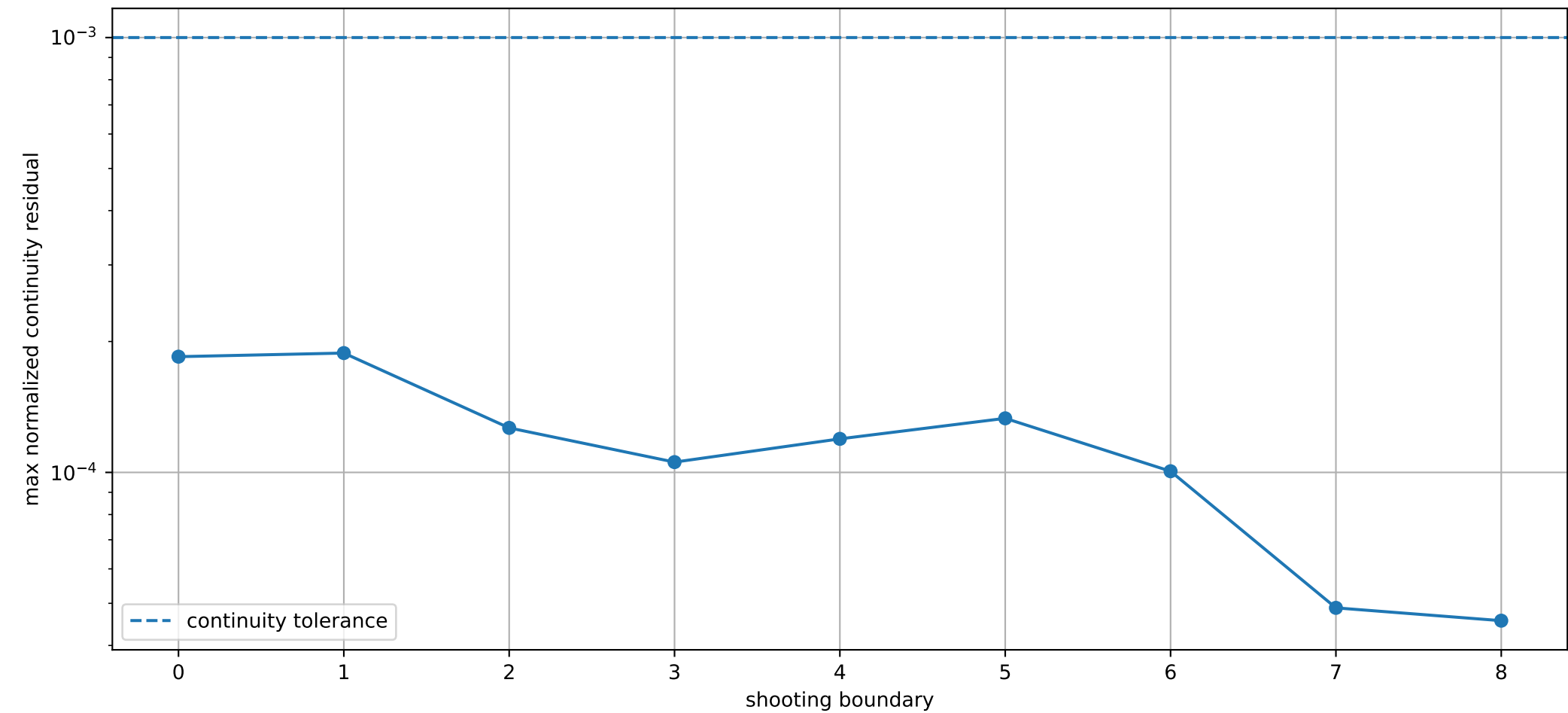
IMU angular velocity (diagnostic; excluded from fit)



IMU specific force (diagnostic; excluded from fit)



Shooting continuity diagnostic



Final estimated parameters

Selected multiple-shooting estimate

Decoded physical parameters

```
mass [kg] 2.59657776509
inertia [kg m^2]
[ 0.0916728079552  0.027365241347 -0.000640582798779]
[ 0.027365241347  0.208957822195 -0.000780210235818]
[-0.000640582798779 -0.000780210235818  0.299644422519]
principal inertia [kg m^2] [0.085601127782, 0.215018948659, 0.299654976228]
CoG offset [m] [0.0226054054618, 0.0105235710798, 0.0106830136579]
rotor force effectiveness [0.856425929502, 1.09686581725, 1.14902040592, 0.926464853266]
command delay [s] 0.16
```

Thirteen optimized smooth coordinates

```
log_mass_scale 0.0990994107626
log_second_moment_cholesky_xx_scale 0.586766658988
log_second_moment_cholesky_yy_scale 0.152833160654
log_second_moment_cholesky_zz_scale 0.00239758797613
normalized_second_moment_cholesky_yx_offset -0.236013698649
normalized_second_moment_cholesky_zx_offset 0.0198116964754
normalized_second_moment_cholesky_zy_offset 0.0391450706718
cog_offset_x_m 0.0226054054618
cog_offset_y_m 0.0105235710798
cog_offset_z_m 0.0106830136579
force_effectiveness_contrast_1 -0.174969543416
force_effectiveness_contrast_2 -0.138947349077
force_effectiveness_contrast_3 0.0881950674834
```

Fit diagnostics

```
stitched inertia-radius loss [m^2] 0.0814507518988
full-rollout inertia-radius loss [m^2] 0.0815200666083
broad soft-prior cost 0.10979678351
position RMSE [m] 0.388443933504
orientation RMSE [deg] 25.8005664123
velocity RMSE [m/s] 1.15010925909
angular velocity RMSE 0.711355165603 rad/s
specific force RMSE 1.16765703351 m/s^2
continuity max (normalized) 0.000188122070606
continuity tolerance 0.001
continuity converged True
```